Offline Policy Evaluation with Real-World Bandit Logs

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Group 7 · DSC6007: Dynamic Programming and Reinforcement Learning

1 Background and research objective

Motivation. Contextual bandits select items to maximize immediate click reward. Online tests expose users to unproven policies; *off-policy evaluation* (OPE) estimates a new policy's value from historical interactions before deployment.

Problem. Only the displayed item's reward is observed. A new policy changes action probabilities, so the logged average reward does not directly estimate its value. Small logging probabilities can give a few observations excessive influence.

Objective. Compare DM, IPS, SNIPS and DR under policy mismatch, and study how sample size and weight clipping affect estimation error, stability and candidate-policy rankings.

Target value: $V(\pi_e) = \mathbb{E}_x\left[\sum_a \pi_e(a \mid x)\,\mathbb{E}[r \mid x, a]\right]$

2 Dataset and preprocessing

26.9 million ZOZOTOWN recommendation records (24–30 Nov 2019), with **Uniform Random** and **Bernoulli Thompson Sampling (BTS)** logs for three campaigns and three display positions.

Campaign	Items	Random test	BTS test
All	80	412,250	3,706,504
Men	34	135,860	1,223,053
Women	46	259,334	2,329,202

Recorded variables. User context x, displayed item a, position, binary click r and logging propensity $\pi_b(a \mid x)$; position is part of the conditioning context. Clicks are rare in the test sets, so uncertainty estimates matter.

Cleaning. Validate actions, clicks and probabilities; remove exact duplicates ($\approx 0.02\%$); align item-category encodings between the two logging policies.

Features and split.

- Chronological split: first 70% of each log for training, last 30% for testing.
- Encoders fitted on Random train only; categories one-hot, affinities $\log(1+\cdot)$ and normalized.
- Context dimensions: 116 / 69 / 80 (All / Men / Women).
- Propensities preserved unclipped for the clipping experiments.

3 Off-policy estimators

Let $\pi_e(a \mid x)$ be the target policy and $\pi_b(a \mid x)$ the logging policy. Define the importance weight w_i, reward prediction q_i and target-averaged prediction m_i:

$$w_i = \frac{\pi_e(a_i \mid x_i)}{\pi_b(a_i \mid x_i)}, \quad q_i = \hat{q}(x_i, a_i), \quad m_i = \sum_a \pi_e(a \mid x_i)\,\hat{q}(x_i, a).$$

Overlap requires $\pi_b(a \mid x) > 0$ wherever $\pi_e(a \mid x) > 0$.

DM : direct method $\hat{V}_{\mathrm{DM}} = \frac{1}{n}\sum_i m_i$

Uses predicted rewards for every action. Low variance, but model misspecification can bias the estimate.

IPS : inverse propensity scoring $\hat{V}_{\mathrm{IPS}} = \frac{1}{n}\sum_i w_i r_i$

Unbiased with correct propensities and action overlap. Extreme weights can create high variance.

SNIPS : self-normalized IPS $\hat{V}_{\mathrm{SNIPS}} = \sum_i w_i r_i \,/\, \sum_i w_i$

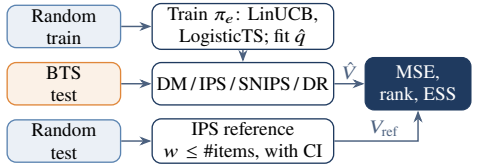
Normalizes total weight and often improves stability, at the cost of finite-sample bias.

DR : doubly robust $\hat{V}_{\mathrm{DR}} = \frac{1}{n}\sum_i \left[m_i + w_i(r_i - q_i)\right]$

Predictions plus weighted residuals. A correct reward model *or* correct propensities suffice, given overlap.

4 Experimental design (planned)

Candidate policies. Train **LinUCB** (uncertainty bonus on predicted reward) and **LogisticTS** (posterior sampling of click-model parameters) on Random train, then freeze them so test data stay separate from learning.



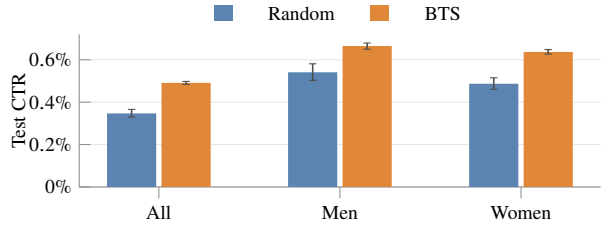
Reference. For each new policy, IPS on independent Random test logs: uniform logging bounds weights by the number of items. The reference has sampling error, so it carries an uncertainty interval.

Sensitivity analysis. Vary sample size and clipping threshold c via $\min(w_i, c)$; repeat subsampling to measure the bias–variance trade-off, monitoring ESS $= (\sum_i w_i)^2 / \sum_i w_i^2$. *Sanity check:* a uniform target policy should reproduce Random CTR.

5 Preliminary results (completed)

Observed test CTR (%) with 95% Wilson intervals; each policy is evaluated on its own test logs.

Campaign	Random	BTS	Lift
All	0.347 [0.330, 0.366]	0.491 [0.484, 0.498]	+41.4%
Men	0.541 [0.503, 0.581]	0.665 [0.650, 0.679]	+22.9%
Women	0.487 [0.461, 0.515]	0.637 [0.627, 0.648]	+30.7%



BTS achieves a relatively **41.4%**, **22.9%** and **30.7%** higher CTR in All, Men and Women. These on-policy benchmarks do not yet determine the most accurate OPE estimator.

6 Evaluation plan and conclusion

Compare DM, IPS, SNIPS and DR across campaigns and policy settings, using repeated subsamples to report error and uncertainty.

Metric	What it measures
MSE	Mean squared error vs. reference
Bias / variance	Systematic error / sampling variability
Spearman rank	Agreement in policy ranking
ESS	Effective sample size under unequal weights

Conclusion so far. BTS beats Random on CTR in all campaigns, but on-policy CTR cannot rank OPE estimators. **Strengths.** Real industrial logs, an independent Random reference, uncertainty throughout. **Limits.** Sparse clicks, weak overlap, propensities down to 10^{-6}. **Outlook.** Finish the MSE, ranking and clipping study; extend to Switch-DR.

References. Saito et al. (2020), *A Large-scale Open Dataset for Bandit Algorithms*; Saito et al. (2021), *Open Bandit Pipeline*; Dudík et al. (2011), doubly robust policy evaluation; Swaminathan & Joachims (2015), self-normalized importance sampling.